Course Code	Name of the Course			
216U03C601	Wireless Communication			
Teaching Scheme	TH	P	TUT	Total
	03	--	--	03
Credits Assigned	03	--	--	03

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites:

- Communication Systems
- Digital Communications

Course Objectives: The objective of this course is to impart knowledge of evolution of wireless communication and fundamental radio concepts, which are at the core of providing mobile communication service to subscribers on the move using limited radio spectrum. Radio propagation losses both large and small describes how to measure and model the impact that signal bandwidth and motion have on the instantaneous received signal through a multipath channel. Second and third generation systems like GSM, CDMA IS95, UMTS are covered. OFDM, MIMO and long-term evolution systems are introduced.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Comprehend fundamentals of mobile communication
CO2. Evaluate various path loss and fading effects
CO3. Compare principles of GSM and IS-95 system.
CO4. Illustrate basics of 3G UMTS technology.
CO5. Summarize enabling technologies for 3GPP LTE-A network

Detailed Syllabus

Module No.	Unit No.	Details	Hrs	CO
1		**Fundamentals of Mobile Communication**	10	CO1
	1.1	Telecom Regulatory Authority of India (TRAI) act & policies		
	1.2	Frequency reuse, and Channel assignment strategies.		
	1.3	Handoff strategies, Practical handoff considerations, Co-channel interference and system capacity, Adjacent channel interference.		
	1.4	Trunking, Grade of Service and related design problems.		
	1.5	**Improving Coverage and Capacity in Cellular System:** Cell splitting, Cell sectoring, A microcell zone concept and Repeaters for range extension.		
2		**Mobile Radio Propagation**	08	CO2
	2.1	**Large Scale Path Loss:** The three basic propagation mechanisms: Reflection, Diffraction, and Scattering.		
	2.2	**Small Scale Fading and Multipath:** Factors influencing small scale fading, and Doppler shift		
	2.3	**Parameters of mobile multipath channels:** Time dispersion parameter,Coherence bandwidth, Dopplerspread and coherence time and related numerical.		
	2.4	**Types of small scale fading:** Fading effects due to multipath time delay spread, Fading effects due to Doppler spread.		
		#**Self-learning topic:** Outdoor Propagation Models: Longley-Rice Model, Okumura Model, Hata Model Indoor Propagation Models: Log-distance path loss model, Attenuation factor model.		
3		**2G/2.5G Technologies**	11	CO3
	3.1	**Global System for Mobile Communication Standard (GSM):** GSM features, services, architecture, channel types, call processing, frame structure and signal processing		
	3.2	**GSM evolution:** GPRS features and network architecture		
	3.3	**Interim Standard-95 (IS-95):** Air interface specifications, channel types, call processing, power control.		
		#**Self-learning topic:** Frequency Division Multiple Access (FDMA, Time Division Multiple Access (TDMA), Direct Sequence Spread Spectrum (DSSS)		
4		**3G Technologies**	04	CO4
	4.1	**Universal Mobile Telecommunication Standard (UMTS):** The WCDMA Air Interface, Objectives, network architecture, UTRAN architecture of UMTS		
	4.2	UMTS Channel types and functions		
	4.3	Codes used in UMTS, Spreading and Scrambling in UMTS		

5	**Enabling Technologies for 3GPP LTE-Advanced Networks**		**12**	**CO5**
	5.1	**Orthogonal Frequency Division Multiplexing (OFDM):** Basic principle of orthogonality, single carrier vs. multicarrier systems, block diagram, cyclic prefix, and Peak to average Power ratio.		
	5.2	**Long Term Evolution (LTE):** Air interface specifications, Architecture and LTE physical layer		
	5.3	LTE-A Enabling Technologies: Carrier aggregation, relaying, and Self-optimizing networks (SON).		
	5.4	Concept of MIMO, spatial diversity, spatial multiplexing, antenna beamforming and space time block codes		
	5.5	Cognitive Radio Networks: Definition and Spectrum sensing		
	5.6	Introduction to 5G network, specifications, applications, use cases, Comparison of 5G and 4G network and applications		
		Total	**45**	

Recommended Books*:

Sr. No.	**Name/s of Author/s**	**Title of Book**	**Name of Publisher with country**	**Edition and Year of Publication**
1.	Theodore S. Rappaport	*Wireless Communications Principles and Practice*	Pearson Publication, India	2nd Edition, 2010
2.	T L Singal	*Wireless Communications*	Tata McGraw Hill, India	2nd Edition, 2017
3.	Guillaume de la Roche, Ben Allen	*LTE-Advanced and Next Generation Wireless Networks-channel modelling and propagation*	Wiley Publication, India	1st Edition, 2013
4.	Young Kyun Kim and Ramjee Prasad	*4G Roadmap and Emerging Communication Technologies*	Artech House Publishers, Boston, London	1st Edition, 2005
5.	Aditya K. Jagannathan	*Principles of Modern Wireless Communication Systems: Theory and Practice*	McGraw-Hill Education, India	1st Edition, 2017
6.	Upena Dalal	*Wireless Communication*	Oxford University Press, India	2nd Edition, 2010